# Contact Finder 🕵️ IN

Enterprise OSINT Multi-Agent AI Engine — Comprehensive Technical Documentation

**Version:** 2.0 (Gemini 3.6 Flash)
**Workspace:** `apps/contact-finder`
**Date:** Sept 14, 2026

### 📌 Project Purpose & Overview

**Contact Finder** is an enterprise-grade multi-agent Open Source Intelligence (OSINT) system designed to track down, attribute, and verify Indian mobile numbers based on names, residential/business addresses, companies, and cities. It is tailored for high-stakes financial recovery (IEPF dividend claims), corporate directory lookups, legal skip tracing, and B2B contact enrichment.

## 1. Multi-Agent Intelligence Core

`CoordinatorAgent.ts`

Master orchestrator. Manages search budgets, batch query execution, corporate entity detection, early exit triggers when confidence $ge 75%$, and fallback social proxy searches.

`ResearcherAgent.ts`

Query expansion & fetching. Generates targeted Google Dorks (combining house numbers, area, city, pincodes) and executes parallel calls across search/B2B APIs.

`ExtractorAgent.ts`

Content parsing engine. Runs 3-pass extraction: Pass-1 Regex, Pass-2 Local Vector Embeddings ( `Transformers.js` ), and Pass-3 LLM context parsing ( `Gemini 3.6 Flash` ).

`VerifierAgent.ts`

Identity resolution & memory. Evaluates name matching, fuzzy address distance ( `Fuse.js` ), domain trust, and maintains the persistent SQLite database cache.

## 2. Cutting-Edge Technical Capabilities

### ⚡ Upgraded LLM Core — Gemini 3.6 Flash

Powered by Google's `gemini-3.6-flash` via `@google/genai` (with OpenRouter Llama 3 fallback). Performs structured JSON context parsing, entity equivalence checking, and relationship resolution without fragile regex rules.

### 🧠 Semantic Vector Embeddings (Transformers.js)

Loads an ONNX-quantized model ( `Xenova/all-MiniLM-L6-v2` ) locally in Node.js. Embeds document paragraphs into mathematical vector space to isolate target contact blocks in 30,000+ line HTML files instantly offline.

### 🕵️ Stealth Puppeteer & Vision Fallback

Uses `puppeteer-extra-plugin-stealth` to spoof WebGL signatures and bypass Cloudflare/Akamai bot protection. If a website obfuscates phone numbers as canvas graphics, it captures a Base64 JPEG screenshot and feeds it to **Gemini 3.6 Flash Vision OCR**.

## 3. Comprehensive Confidence Scoring & Attribution Formula

**Total Score (0 - 100%) =**
`S_NameMatch + S_Attribution + S_AddressMatch + S_LocalitySnippet + S_MultiSourceBonus + S_DomainTrust`

### Attribution Types & Decision Matrix

| Attribution Type | Points | Description & Context Match | Tag |
|---|---|---|---|
| `DIRECT_PERSON` | `+20` | Direct line belonging to the subject or their father. | `self` / `father` |
| `PROXY_PERSON` | `+10` | Verified colleague, co-director, or associate at same company. | `proxy` / `coworker` |
| `RWA` | `+5` | Resident Welfare Association official or Society Secretary. | `rwa_official` |
| `BROKER` | `+5` | Real estate agent or property dealer handling the premises. | `broker` |
| `ORGANIZATION` | `+3` | Corporate office landline or reception desk. | `organization` |
| `GENERIC` | 0 | Header lines ("Contact Us") without specific entity names. | `generic` |
| `AMBIGUOUS` | -5 | Multiple conflicting names listed nearby. | `ambiguous` |
| `UNKNOWN` | -10 | Orphaned phone digits extracted without context. | `unknown` |

### Scoring Breakdown Components

- **Name Match (0 to +30 pts):** Exact Match (+30), Variant/Middle Name (+20), Surname Only (+10), Partial (+5).
- **Address Match (0 to +40 pts):** Exact House/Flat Number Match (+40), Area/Colony Match (+20), City Match (+5), Snippet Locality (+10).
- **Multi-Source Verification (+15 pts per domain):** $15 imes (N - 1)$ bonus points for $N$ independent web domains discovering the same number.
- **Domain Trust (+5 to +10 pts):** Government portals ( `.gov.in` , `mca.gov.in` ) (+10), Established B2B directories ( `ZaubaCorp` , `IndiaMart` , `TradeIndia` ) (+5).

## 4. Integrated OSINT Data Providers

| Category | Providers Integrated | Primary Source File |
|---|---|---|
| **Search Engines** | SerpAPI (Google), Perplexity AI (Sonar Pro), TinyFish | `SerpAPIProvider.ts` , `PerplexityProvider.ts` |
| **Corporate Registries** | ZaubaCorp, QuickCompany, TheCompanyCheck, Explorium, Apify Zauba | `ZaubaCorpScraper.ts` , `ExploriumProvider.ts` |
| **B2B & Local Directories** | IndiaMart, TradeIndia, JustDial | `IndiaMartProvider.ts` , `JustDialProvider.ts` |
| **LLM & Vision** | Google Gemini 3.6 Flash, OpenRouter (Llama 3) | `GeminiService.ts` , `OpenRouterService.ts` |
| **Caller ID Services** | Truecaller Lookup Proxy & Internal Knowledge Database | `CallerIdService.ts` , `DatabaseService.ts` |

## 5. Core Features & UI Applications

### 1. Single Investigation UI ( `/` )

Interactive search dashboard. Streams live OSINT progress, displays confidence badges, relationship tags, and complete evidence chains.

### 2. Bulk CSV Engine ( `/bulk` )

Processes datasets with hundreds of shareholder folios concurrently. Features styled Excel exports ( `exportExcel.ts` ) with hyperlinked sources.

### 3. PDF Parser & Converter ( `/pdf-converter` )

Parses legal records, dividend filings, and scanned PDFs using `pdf-parse` , `pdf2json` , and `tesseract.js` OCR directly into investigation pipelines.